Practical - 4

Practical Title: JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Problem Definition: Add dynamic UI components such as collapsible FAQ, modal popup, image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript.

A. Theory

1. How are DOM elements selected and modified?

Answer:

DOM elements are selected using JavaScript document methods like querySelector() and modified by updating their properties, styles, or structural methods. This process forms the foundation of dynamic web development

https://www.geeksforgeeks.org/javascript/how-to-select-dom-elements-in-javascript/

2. Are event listeners attached correctly?

Answer:

Yes. Event listeners are attached using addEventListener(). In our website, event listeners are used for actions such as the theme toggle and registration form submission. This allows the webpage to respond to user actions without putting JavaScript directly inside HTML elements.

Reference: W3C – Web Programming Guidelines

3.Is local Storage used for remembering theme choice?

Answer:

Yes ,local storage is widely used by websites to remember user choices and preferences

https://cookie-script.com/blog/local-storage-and-session-storage

4.Does interactivity improve usability without breaking accessibility?

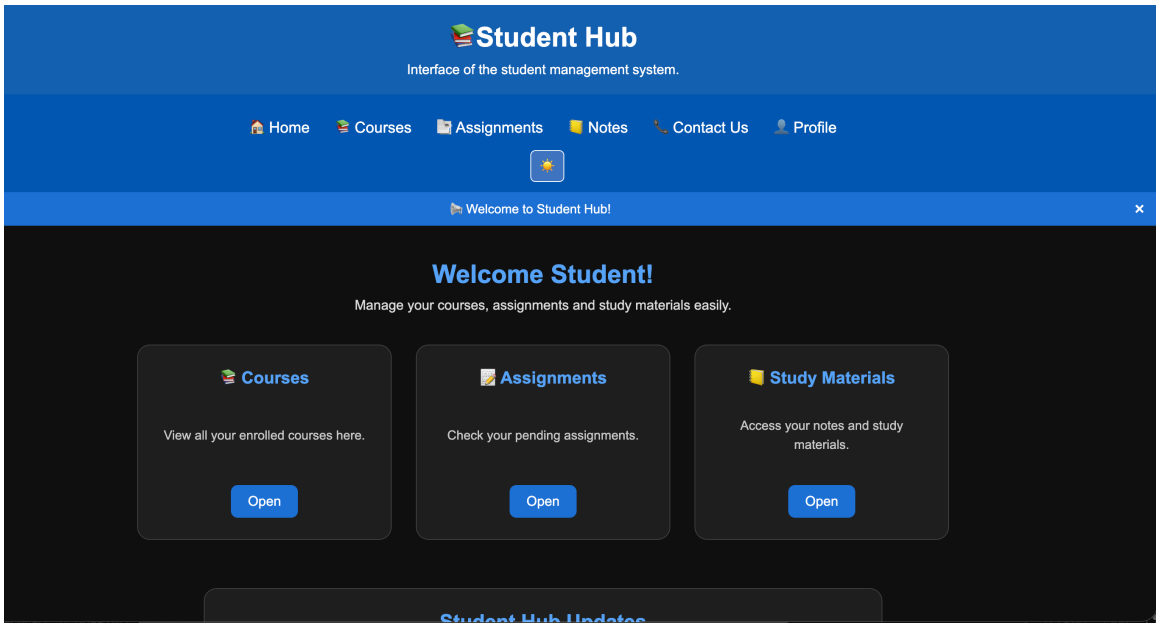
Answer:

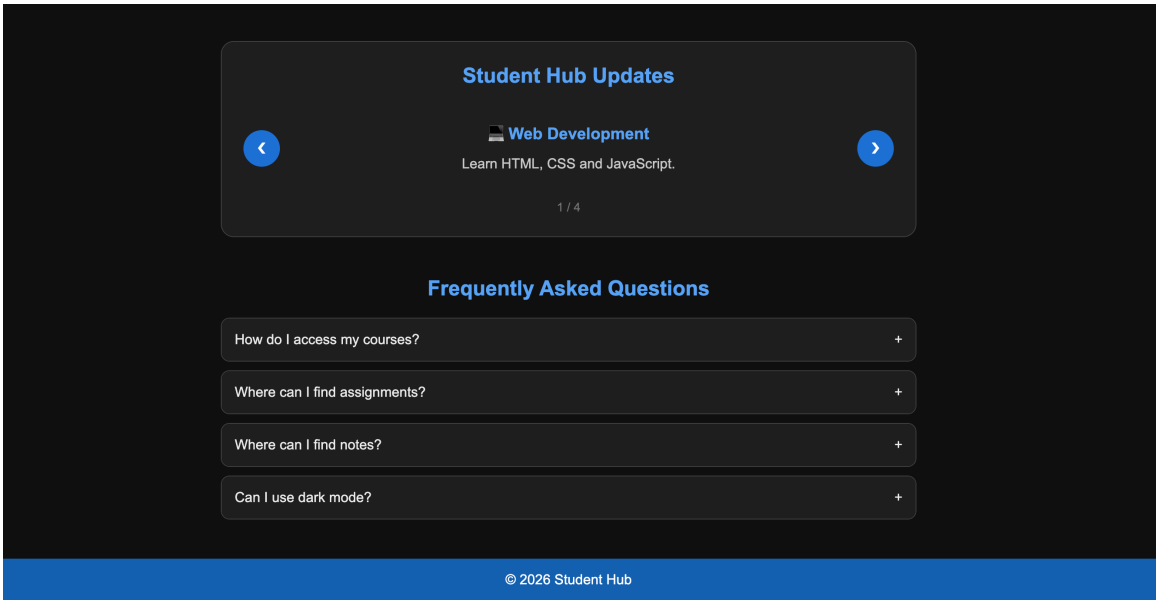
Yes, well-designed interactivity improves overall usability without breaking accessibility, provided interactive components are built to support assistive technologies and standard navigation rules.

B. Result

The StudentHub pages were successfully changing its theme from dark mode to light mode perfectly without any concerns

C. Evidence of Work (Screenshots)





D. GitHub Repository Link:

https://github.com/Dhanraj2408/StudentHub

E. Challenges Faced:

When I made the dark and light mode the code and all were perfect but still button was not working at all as if it was unclickable then I added pop up notification with button as button is clicked then it started working .

F. Learning Outcomes:

After this practical I learn how to use java script toggle the screen add drop down menu and etc.

Faculty Remarks: